Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	26 June 2026
Team ID	Solo
Project Name	ShopEZ - E-Commerce Application
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

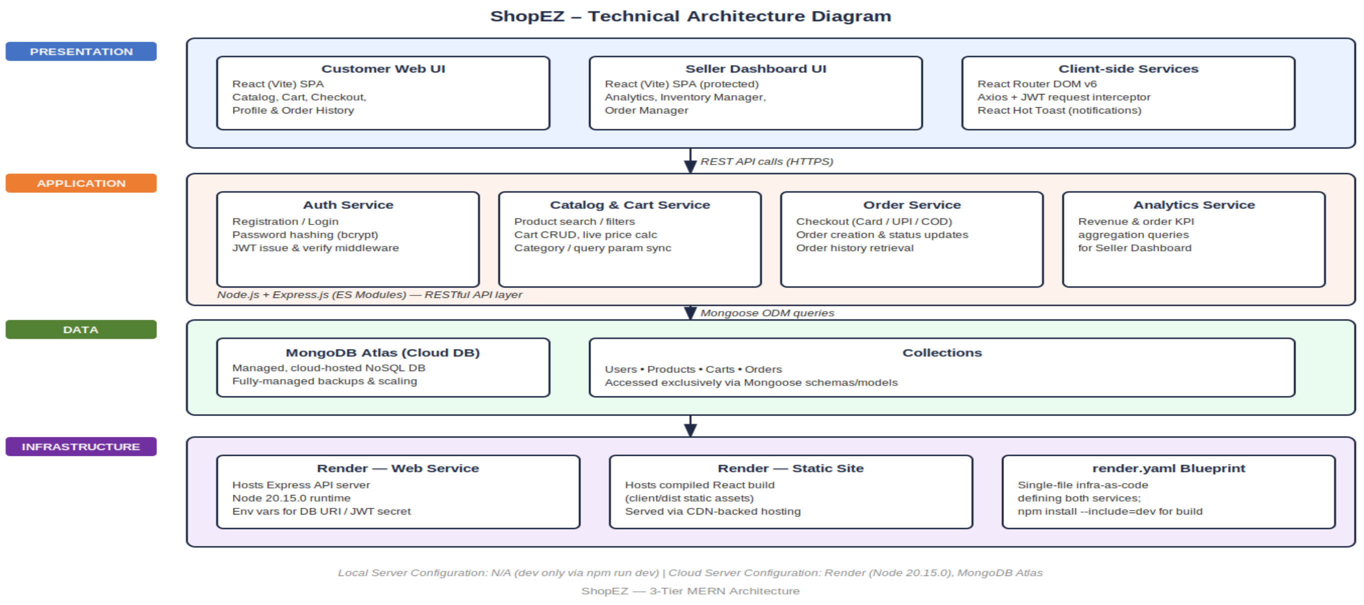

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Web UI for customers to browse/shop and for sellers to manage their store	React (Vite), React Router DOM v6, React Icons, React Hot Toast
2.	Application Logic-1	Core business logic — auth, catalog, cart, checkout, order processing	Node.js, Express.js (ES Modules)
3.	Application Logic-2	Seller-side logic — inventory CRUD and order/status management	Node.js, Express.js REST controllers
4.	Application Logic-3	Seller analytics logic — aggregating revenue and order counts for KPI cards	Express.js + MongoDB aggregation via Mongoose
5.	Cloud Database	Managed, cloud-hosted database service	MongoDB Atlas
6.	File Storage	N/A for current scope (no dedicated file/image upload service in use)	Not applicable / Local filesystem (static assets)
7.	External API-1	Not applicable — no external API dependency in current scope	Not applicable
8.	External API-2	Not applicable — no external API dependency in current scope	Not applicable
9.	Machine Learning Model	Not applicable — no ML model used in current scope	Not applicable
10.	Infrastructure (Server / Cloud)	Application deployed entirely on cloud, no local server dependency. Local Server Configuration: N/A (development only, via npm run dev). Cloud Server Configuration: Render web service (Node 20.15.0) running the Express server + static site for the React build	Render (Cloud PaaS), unified render.yaml blueprint

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Frontend and backend both built entirely on open-source frameworks	React, Express.js, Mongoose, Node.js
2.	Security Implementations	Stateless authentication with signed tokens; passwords never stored in plain text; secrets excluded from version control; purged exposed credentials from Git history	JWT (JSON Web Tokens), bcrypt password hashing, .gitignore for .env secrets
3.	Scalable Architecture	3-tier architecture (presentation, application, data) with a clear separation between client/ and server/, allowing each layer to scale or be redeployed independently	3-tier MERN architecture (React – Express – MongoD
4.	Availability	Single-blueprint deployment keeps client and server in sync and simplifies redeploys; MongoDB Atlas provides managed uptime/replication for the database layer	Render unified deployment + MongoDB Atlas managed hosting
5.	Performance	Vite-based build produces optimized, minified static assets for fast load times; Axios interceptor avoids redundant auth calls by reusing the stored JWT; Mongoose queries are scoped/indexed on key fields (e.g., category, order status) for fast catalog and order lookup	Vite build optimization, Axios request interceptors, MongoDB indexing via Mongoose